

Helping Teachers Use Google Classroom for Effective Formative Assessment

A scenario-based faculty development module | Articulate Storyline 360

Developer	Trina Dutta
Tool	Articulate Storyline 360 (Trial)
Audience	School teachers (K–12) using Google Classroom
Slides	18 slides approximately 30 minutes
Completed	July 2026

1. Project Overview

This case study documents the instructional design and Storyline 360 development of an 18-slide, 30minute scenario-based eLearning module for school teachers. The module was designed using the ADDIE model and addresses a real performance gap: teachers who use Google Classroom primarily as a file repository, without leveraging its formative assessment features. The project was built independently as a portfolio piece demonstrating instructional analysis, scenario-based design, adult learning principles, and professional Storyline 360 development.

2. The Learning Problem

Teachers regularly using Google Classroom uploaded assignments and shared files but did not use features such as the Question tool, Quiz Assignment, Google Forms integration, Comment Bank, or the Student Responses interface for formative assessment purposes.

Consequences of this gap included:

- Feedback was delayed — often returned days after the learning moment had passed.
- Teachers had limited real-time data on student understanding during instruction.
- Google Classroom's most powerful formative features remained unused.
- Students received less frequent, less targeted, and less timely feedback.

3. Business Need

Schools investing in Google Workspace for Education need measurable returns in the form of improved teacher practice and student outcomes. The specific need was for teachers to provide faster, more targeted formative feedback while reducing the administrative burden of manual notebook checking — achievable within a short, practical, self-paced module.

4. Learner Analysis

Learner Profile

Characteristic	Description
Role	School teachers, primary and secondary (Grades 4–12)
Digital literacy	Beginner to intermediate — comfortable with email and file sharing; less confident with interactive digital tools

Google Classroom experience	Currently use it to post and receive assignments
Prior knowledge	Aware of formative assessment conceptually; limited confidence in digital implementation
Time available	Limited — module must be completable in a single 30-minute sitting
Characteristic	Description
Motivation	Intrinsically motivated to help students; may also be required by school PD policy

Learner Persona

Meet Ms Priya — The Module Protagonist

Ms Priya teaches Grade 7 Science at an urban school. She has 9 years of experience and uses Google Classroom daily to share PDFs and collect assignments. On weekends she manually checks 40 notebooks to gauge understanding. She knows there must be a faster way but has never had time to explore it. Ms Priya serves as both the instructional scenario anchor and the narrative protagonist throughout the module, appearing in 8 of 18 slides.

5. Learning Objectives

By the end of this module, learners are able to:

1. Explain the purpose and characteristics of formative assessment.
2. Differentiate between formative and summative assessment using examples.
3. Select the appropriate Google Classroom feature for different classroom situations.
4. Create a Question activity in Google Classroom.
5. Create a Quiz Assignment in Google Classroom.
6. Review student responses using the Google Classroom responses interface.
7. Provide effective, specific, and timely digital feedback to students.

6. Instructional Strategy

Why Scenario-Based Learning?

Scenario-based learning (SBL) was selected as the primary strategy because the performance gap is fundamentally a decision-making gap — knowing which feature to use in which situation — not merely a knowledge deficit. SBL allows learners to practise judgment in realistic, low-stakes situations before applying skills in their classroom. Research supports SBL as more effective than expository instruction for transfer of learning to job performance (Clark & Mayer, 2016).

The module uses Ms Priya as a narrative anchor across all 18 slides. Learners first observe her situation, then make decisions alongside her, and finally apply their own judgment in the Final Challenge.

Adult Learning Principles Applied

Principle	Application
Need to know	Module opens with Ms Priya's real problem — establishes immediate relevance before any content is introduced.
Principle	Application
Self-concept	Non-punitive scoring throughout. Retry always available. Learners make their own decisions in all interactive activities.
Prior experience	Reflection prompts invite learners to connect content to their own classrooms. Scenario situations reflect real teacher challenges.
Readiness to learn	Content directly applicable to Monday's class — no abstract theory without classroom application.
Orientation to learning	Problem-centred, not subject-centred. Every screen addresses: what would I do in this situation?
Motivation	Intrinsic framing — this helps your students. Module closes with a personal commitment reflection.

7. Module Structure — 18 Slides As Built

Slide	Title	Purpose & Key Feature
01	Welcome & Title Screen	Navy/teal title screen with Start button. Establishes visual identity.
02	How to Navigate	4-card navigation guide. Sets learner expectations.
03	Meet Ms Priya	Character introduction with speech bubble. Narrative hook.
04	Learning Objectives	7 objectives with Marilyn character. Schema activation.
05	What Is Formative Assessment?	Two-card definition + reflection prompt. Objective 1.
06	Formative vs Summative	Click-to-reveal 4 statement cards. Objective 2.
07	Explore Google Classroom Features	5-button feature menu with layers. Objective 3.

08	Feature Deep Dive — Question Tool	Two-panel content slide with Ms Priya scenario. Objective 3.
09	Guided Demo — Question Activity	5-step layered demo with Marilyn at desk. Objective 4.
10	Guided Demo — Quiz Assignment	5-step layered demo (duplicate structure). Objective 5.
11	Scenario 1 — Fractions Check	Branching scenario with correct/incorrect feedback. Objective 3.
12	Scenario 2 — Group Project	Branching scenario — Google Forms. Objective 3.
13	Scenario 3 — Weekly Comments	Branching scenario — Comment Bank. Objective 3.
14	Scenario 4 — Auto-	Branching scenario — Quiz Assignment. Objective 3.
Slide	Title	Purpose & Key Feature
	Graded Test	
15	Role-Play — Student Responses	3-branch consequence role-play. Objectives 6 & 7.
16	Final Challenge — Your Mission	3-step synthesis builder on navy background. All objectives.
17	Key Takeaways	5 click-to-reveal takeaway cards. Consolidation.
18	Course Complete	Reflection text entry + Exit course. Closure.

8. Storyline Features Used

Feature	Where Used	Why It Supports Learning
Layers	Slides 07, 09, 10, 11–15, 16 — feature reveals, step demos, scenario feedback, mission builder	Allow complex content on one slide without excessive slide count. Keep related content contextually connected.
Show/Hide Layer Triggers	Slides 07, 09, 10, 11–15, 16	Connect learner actions to instructional consequences — the foundation of all branching interactivity.
States (Custom & Built-in)	Slide 06 (4 cards), Slide 17 (5 takeaway cards)	Provide visible feedback on interaction progress. Built-in states (Visited, Selected, Drop Correct) used as trialversion workaround.

Jump to Slide Triggers	All Continue buttons, scenario Next buttons	Controls module flow and narrative progression across slides.
Exit Course Trigger	Slide 18 Finish button	Professional module closure — signals completion to LMS if deployed.
Content Library Characters	Slides 03, 04, 08, 09, 10 — Marilyn in standing and seated-at-desk poses	Marilyn at desk on demo slides adds authenticity to simulation context. Standing pose used for narrative slides.
Text Entry Input	Slide 18 reflection box	Low-stakes personal commitment activity. Respects adult autonomy — response is private, not graded.
Button Triggers	All slides	Primary navigation mechanism. Player Next/Prev buttons disabled — all navigation via custom Continue buttons.

9. Known Limitations & Design Decisions

Trial Version Adaptations

- Drag-and-drop on Slide 06 replaced with click-to-reveal states — same conceptual learning outcome achieved with a simpler mechanic.
- Final Challenge (Slide 16) uses a fixed Mission Complete result rather than personalised variable-based scoring — all paths reveal the ideal answer for self-comparison.
- Progress bar mentioned in Slide 02 navigation guide replaced with a 'Slides' information card — honest about module structure without requiring complex variable-driven progress tracking.

Design Simplifications

- Marilyn removed from Role-Play branch layers (Slide 15) after testing showed character at original scale conflicted with consequence text readability. Text-only consequence cards retained.
- Slide count reduced from planned 19 to 18 — the Formative vs Summative Results slide was merged into the click-to-reveal interaction on Slide 06, reducing redundancy.

10. Visual Design

The module uses a consistent three-colour palette: navy (#1A2B5F), teal (#0D7377), and light grey (#F2F4F7). Navy is used for title and completion screens to bookend the module. Teal is used for all interactive elements, header bars, and emphasis. Light grey is the standard content slide background.

All fonts are Arial throughout. Minimum body text size 14pt on screen. Teal on white contrast ratio 4.6:1 (WCAG 2.1 AA compliant). Navy on white 8.6:1.

Content Library character Marilyn (illustrated style) is used consistently across 8 slides in two poses: standing/neutral for narrative slides and seated-at-desk for guided demonstration slides.

11. Portfolio Reflection

This module represents my first independently designed and developed eLearning project in Articulate Storyline 360. Building it taught me that instructional design decisions and technical development decisions are inseparable — every choice about triggers, layers, and states had to serve a learning purpose, not just demonstrate a technical skill.

The most significant design decision was choosing scenario-based learning over a content-presentation format. After analysing the performance gap, I concluded that teachers were not using Google Classroom's features because they lacked confidence in applying them in context — not because they lacked awareness. Scenarios simulate the judgment calls teachers need to make in real classrooms, which a quiz-only or lecture format cannot replicate.

Working within the constraints of the trial version strengthened my problem-solving skills. When drag-and-drop was unavailable, I used states creatively to achieve the same instructional outcome. When variables were complex to implement, I designed a self-reflective completion experience instead of an automated scoring system. These adaptations are documented as known limitations — a professional practice that demonstrates honest, realistic instructional design thinking.

Ms Priya as a narrative device was one of the most effective design choices. By threading a single character with a relatable problem across all 18 slides, I created emotional continuity and a consistent frame of reference for every interaction. Learners are not just answering questions — they are helping Ms Priya solve a real problem they recognise from their own classrooms.

This project demonstrates that I can move from problem analysis to a fully published, interaction-rich eLearning module — the complete workflow of professional instructional design practice.